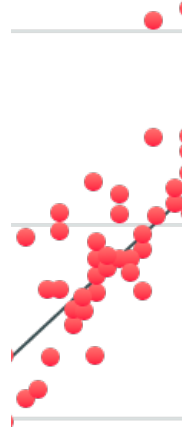
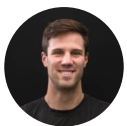


October 9, 2024

Validity and Reliability of a Mobile Application (Metric) for Velocity-based Training (VBT) in the Bench Press Exercise



We're excited to share the results of a recent validation study on Metric VBT, conducted by Antonio Šagovac as part of his Master's studies at the University of Zagreb.



Jacob Tober
Co-founder

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We're excited to share the results of a recent validation study on Metric VBT, conducted by Antonio Šagovac as part of his Master's studies at the University of Zagreb.

This study, focused on the bench press exercise, provides compelling

evidence for Metric VBT's accuracy and reliability when compared to another commercial velocity measurement tool.

This article is written with thanks to Antonio who was kind enough to share the results and a copy of his paper with us. He has allowed us to translate the original paper into English which you can find linked here.

[You can read the original paper written in Croatian here →](#)

[Or the English translation here →](#)

Any errors in translation are mine and not Antonio's.

Study details

Lead author: Antonio Šagovac

Title: Validity and Reliability of a Mobile Application (Metric) for Velocity-based Training (VBT) in the Bench Press Exercise

Date of publication: September, 2024 (Master's thesis)

Metric version: v4.5.0

TL;DR: Metric is reliable and valid when compared to commercially available Linear Positional Transducer Vitruve on the Bench Press.

Metric version 4.5 is a valid and reliable velocity tracking solution

This paper, marks the first validation of a post-beta version of the Metric VBT tracking algorithm.

While earlier validations of Metric with [our internal validation study](#), and external validations of [v0.5.4 by Chris Taber](#) published in November

2023, and then July 2024 with [a v0.6.0 validation by Danielle Trowell](#) showed mixed validity and reliability, much work has gone into Metric since these studies completed their data collection. Thanks in large part to continued feedback from lifters using Metric in their training, our development team has shipped a major rewrite to our tracking algorithm in November 2022 with version 1.0, and ongoing iterative improvements ever since.

These findings validate our continuous improvement efforts and align with user feedback: Metric VBT has matured into a reliable, accurate tool for real-world training. We're excited to see scientific validation supporting what our users already experience – a powerful, user-friendly VBT solution that consistently delivers in practical settings.

Study Overview

Before diving into the results, here is a quick summary of the context for this research:

- **Comparison Study:** This research compared Metric VBT (v4.5.0) to the Vitruve linear position transducer (LPT) device, another commercial product used for velocity-based training.
- **Focus on Bench Press:** The study specifically examined the bench press exercise, a fundamental movement in resistance training.
- **Participant Profile:** 15 male participants with at least 2 years of resistance training experience were involved in the study.
- **Load Ranges:** Measurements were taken at 45%, 60%, and 75% of participants' one-repetition maximum (1RM) across 150 total repetitions.

It's worth noting that while 3D motion capture is often considered the gold standard for movement analysis, this study focused on comparing two practical, and commercially available tools that coaches and athletes are likely to use in real-world training environments.

Key Findings

The results of this study are extremely encouraging for Metric VBT users:

Metric showed high validity for all measures, correlating strongly with the Vitruve LPT device

- Mean Velocity: $r = 0.93$ (almost perfect correlation)
- Peak Velocity: $r = 0.91$ (almost perfect correlation)
- Range of Motion: $r = 0.75$ (very high correlation)

These correlations indicate that Metric VBT's measurements closely align with those of the Vitruve LPT device across all measured variables.

Reliability
of Metric
VBT vs
Vitruve
LTP –
Mean
velocity

Reliability
of Metric
VBT vs
Vitruve
LTP –
Peak
velocity

Reliability
of Metric
VBT vs
Vitruve LTP
– Range of
Motion.
Note that
Vitruve
does not
apply an
X-axis
correction
while the
Metric
application
does

Excellent Reliability & low variability across the different loads and velocities used in the study

- Intraclass Correlation Coefficients (ICC) ranged from 0.89 to 0.97 across all variables and loads.

- Coefficient of Variation (CV) values were below 10% for all measurements.

This demonstrates excellent consistency in Metric VBT's measurements across multiple repetitions, different users and on different absolute bar speeds and loading intensities.

Metric VBT performed best on the 60% and 75% of 1RM intensities (lower absolute bar speeds), showing high correlations and reliability. While performance at 45% 1RM was good, it was slightly less consistent than at higher loads.

A low overall systematic error, that could be attributed to differences in raw data processing and not raw accuracy of the systems.

- Mean Velocity: Slight overestimation (0.01 ± 0.06 m/s)
- Peak Velocity: Slight underestimation (-0.08 ± 0.12 m/s)
- Range of Motion: Slight overestimation (4.38 ± 4.14 cm)

These small systematic errors are within acceptable ranges for practical use and likely attributable to differences in raw data processing between Vitruve and Metric.

The Vitruve LPT measures only vertical displacement making device placement a key factor in reliability and giving Vitruve a potential limitation when tracking lifts with a curved bar path such as bench press and power cleans. Metric accounts and corrects for motion in the X, Y and Z axes during repetitions, calculating barbell velocity and range of motion across a range of device placements and on curved exercises or purely vertical movements alike.

While high validity is always desirable, for coaches and lifters looking to

rely on this technology in their training, reliability (consistency) of your tracking technology is the most important factor of all.

Something the study's author points out:

These findings suggest that despite the existence of systematic error and moderate to high correlation at all loads in some variables, the mobile application becomes practically justified thanks to very high levels of reliability. In real conditions, reliability becomes a much more important characteristic than validity because if the measuring instrument consistently shows the same systematic error, while being consistently consistent in terms of results, then it is possible to track the trend of changes if the training process is carried out under the same conditions.

This study shows that compared to Vitruve, Metric shows low systematic error and can be relied upon to deliver consistent and useful training data.

The inverse of this statement is also true: compared to Metric, Vitruve shows low systematic error and can be relied upon to deliver consistent and useful training data.

Metric is a reliable and valid tool to use in your training

This study provides strong evidence that Metric VBT is a robust and useful tool for measuring velocity and range of motion in the bench press exercise. By demonstrating high correlations with a dedicated LPT device and excellent reliability across repetitions, this research confirms that Metric VBT can be trusted for accurate, consistent measurements in your training.

Ultimately, the mobile application satisfies a high level of reliability for individual repetitions and acceptable variability of results, making it metrically justified and safe to use.

It's important to note that this study is a comparison of two technologies rather than a validation against a gold standard like 3D motion capture. Both Metric and Vitruve have their own strengths and limitations. However, the results clearly show that Metric is capable of providing professional grade velocity tracking on par with those of a commercially available LPT device, making it a viable and more accessible alternative for velocity-based training.

[You can download the Metric VBT app for iOS at this link →](#)

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By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



METRIC

Barbell velocity tracking

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